DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2021/2022

YEAR III SEMESTER II SUPPLEMENTARY UNIVERSITY EXAMINATIONS FOR THE DEGREE OF BACHEROR OF SCIENCE IN INFORMATION TECHNOLOGY

CIT 3104: COMPUTER GRAPHICS

DATE: March 2022 **TIME: 2 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

a) Giving and example in each case, explain the following terms as used in computer graphics
   i. Interactive Computer Graphics
   ii. Passive Computer Graphics (4 marks)

b) Using relevant examples, briefly discuss any four application areas of computer graphics. (8 marks)

c) Colour display CRT monitors display colour pictures through combination of phosphorous that emits different colour light. Outline the two techniques for producing colour displays with CRT.(2mks)

d) What are the three major components of a presentation graphics application? (3marks)

d) Graphic illustrations are used in presentations to help convey an idea or express a mood. Distinguish between **vector** and **Bitmap** illustrations graphics giving one advantage and one disadvantage of each. (4 marks)

e) Animation adds to graphics the dimension of time which vastly increases the amount of information to be transmitted. Historical approach of animation has classified computer animation into two main categories. Briefly describe the two categories. (4 marks)

e) What is scan conversion? (2marks)

e) Identify the labelled parts of the sample OpenGl function command shown below. (3mks)

```
glColor3f(1.0, 1.0, 1.0);
```

Question 2: (20 marks)

a) Outline the basic steps involved in converting Cartesian coordinate system to the system understandable points. (6marks)

b) Differentiate between random scan and raster scan monitors . (4mks)

c) Explain each of the following OpenGl geometric primitive names as used in OpenGL programs. (10mks)

GL_POINTS
GL_LINES
GL_LINE_STRIP
GL_LINE_LOOP
GL_TRIANGLES
GL_TRIANGLE_STRIP
GL_TRIANGLE_FAN
GL_QUADS
GL_QUAD_STRIP
GL_POLYGON

Question 3: (20 marks)

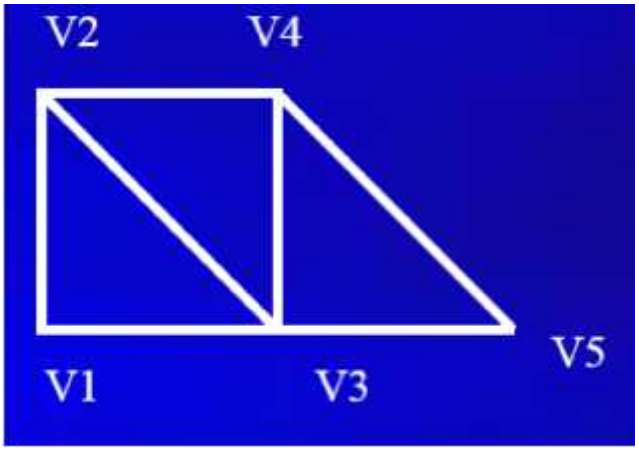
a) Differentiate between Quadtree and Octree data structures as used in computer graphics. (4mks).

b) Explain the function of each of the following OpenGl functions as used for drawing primitives (4marks)

i) glBegin(mode);
ii) glColor3f(1,0,0);
iii) glVertex3f(0,1.5,-2);
iv) glEnd(void);

c) Write a program code snippet that generates a GL triangle_strip such as one shown below.

(8marks)



d) Using a well labelled diagram, explain how a refresh CRT display works. (4mks)

Question 4: (20 marks)

a) Most modern computers are fitted with a dedicated graphics processor to perform the various tasks that relate to computer graphics. Identify and explain any such five tasks. (5marks)

b) Explain the following terms as used in 3D graphics models. (6marks)
i) Rendering
ii) Ray tracing
iii) Texture mapping
iv) Pixel-value

c) Describe the following 3D display techniques. (6mks)

i) Perspective projection

ii) Parallel projection

iii) Cueing

d) Even though the rasterization tries to generate a completely straight line, yet in few cases we may not get equal brightness. Explain any three compromises that are made while generating such lines. (3mks)

Question 5: (20 marks)

a) Consider two raster systems with resolutions 1024 by 600 and 1366 by 768.

i) What is the standard Aspect Ratio in use by the two raster systems described above? (4marks)

ii) Determine the number of pixels that should be accessed per second if the display controller refreshes the screen at a rate of 30Hz. (2mks)

iii) Determine the number of pixels that should be accessed per scan cycle if the system uses the interlacing technique. (2mks)

iv) Determine the minimum difference in storage capacity (per second) between the two systems if each system is defined using 24 bits. Give your solution in kilobytes. (4mks)

b) i) what is a computer graphics model? (2mks)

ii) Giving an example in each case, discuss any three types of models that can be implemented using computer graphics. (6mks)